

8 (a) (i) probability of decay (of a nucleus) <u>per</u> unit time	M1 A1 [2]
(ii) $\lambda t_{1/2} = \ln 2$ $\lambda = \ln 2 / (3.82 \times 24 \times 3600)$ $= 2.1 \times 10^{-6} \text{ s}^{-1}$	M1 A0 [1]
(b) $A = \lambda N$ $200 = 2.1 \times 10^{-6} \times N$ $N = 9.5 \times 10^7$ ratio = $(2.5 \times 10^{25}) / (9.5 \times 10^7)$ $= 2.6 \times 10^{17}$	C1 C1 A1 [3]

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Section B

0 (c) any value greater than, or equal to, 5 kΩ

B1 [1]